

P1

SIMPLE LINEAR MODEL

$$y_i = \beta_0 + \beta_1 \cdot x_i + \varepsilon_i \quad ; \quad \text{var}(\varepsilon) > 0$$

①

$$y_i = \beta_0 + \beta_1 \cdot x_i + \varepsilon_i$$

$$\Rightarrow \varepsilon_i = y_i - \beta_0 - \beta_1 \cdot x_i$$

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i)^2$$

$$\left\{ \hat{\beta}_0 \right\} : 2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i) \cdot (-1) = 0$$

$$= - \sum_{i=1}^n y_i + \sum_{i=1}^n \hat{\beta}_0 + \hat{\beta}_1 \cdot \sum_{i=1}^n x_i = 0$$

$$\therefore n \hat{\beta}_0 = \sum_{i=1}^n y_i - \hat{\beta}_1 \cdot \sum_{i=1}^n x_i$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \cdot \bar{x}$$

WE ARE MINIMIZING:

$$\begin{aligned} &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i)^2 \quad \text{WITHOUT REPLACEMENT} \\ &= \sum_{i=1}^n (y_i - \bar{y} + \hat{\beta}_1 \cdot \bar{x} - \hat{\beta}_1 \cdot x_i)^2 \quad \Rightarrow \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i) \cdot x_i = 0 \\ &= \sum_{i=1}^n (y_i - \bar{y} - \hat{\beta}_1 (x_i - \bar{x}))^2 \end{aligned}$$

F.O.C :

$$\left\{ \hat{\beta}_1 \right\} : 2 \cdot \sum_{i=1}^n (y_i - \bar{y} - \hat{\beta}_1 (x_i - \bar{x})) \cdot (-1) \cdot (x_i - \bar{x})$$

$$\Rightarrow - \sum_{i=1}^n (y_i - \bar{y}) \cdot (x_i - \bar{x}) + \hat{\beta}_1 \cdot \sum_{i=1}^n (x_i - \bar{x})^2 = 0$$

$$\therefore \hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y}) (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

NOTE: YOU CAN ALSO WRITE $\hat{\beta}_1$ AS:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) \cdot y_i - \sum_{i=1}^n (x_i - \bar{x}) \cdot \bar{y}}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{BUT } \sum_{i=1}^n (x_i - \bar{x}) \cdot \bar{y}$$

$$\begin{aligned} \therefore \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x}) \cdot y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{AND } \sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} \\ &= \sum_{i=1}^n x_i - n \cdot \bar{x} \\ &= \sum_{i=1}^n x_i - \frac{n}{n} \sum_{i=1}^n x_i \\ &= 0 \quad \dagger \end{aligned}$$

TUTORIAL 3

① WE OBSERVE AN IID SAMPLE $\{y_i, x_i\}_{i=1}^n$

NOTE: IID $\rightarrow$ INDEPENDENT: KNOWING ONE OBSERVATION TELLS YOU NOTHING ABOUT ANOTHER.

$\rightarrow$ IDENTICALLY DISTRIBUTED: EVERY OBSERVATION COMES FROM THE SAME PROBABILITY DISTRIBUTION

IN THE SAMPLE, OLS CHOOSES $(\hat{\beta}_0, \hat{\beta}_1)$ TO MINIMIZE THE SUM OF SQUARED RESIDUALS

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

WE DEFINE FITTED VALUES AND RESIDUALS:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$$u_i = y_i - \hat{y}_i$$

MECHANICALLY YOU WILL OBTAIN THE FOLLOWING FROM NORMAL EQUATIONS (LAST TUTORIAL).

② ORTHOGONALITY PROPERTIES

① RESIDUALS SUM TO ZERO

FROM (1)

$$= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$$

$$= \sum_{i=1}^n u_i = 0 \rightarrow \text{RESIDUALS AVERAGE TO ZERO.}$$

③ RESIDUALS ARE ORTHOGONAL TO $(x_i - \bar{x})$

$$\sum_{i=1}^n (x_i - \bar{x}) u_i = \sum_{i=1}^n x_i u_i - \sum_{i=1}^n \bar{x} u_i$$

$$= 0 - \bar{x} \cdot \sum_{i=1}^n u_i = 0$$

same interpretation

② FROM (2): RESIDUALS ARE ORTHOGONAL TO REGRESSORS

$$= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \cdot x_i = 0$$

$$= \sum_{i=1}^n u_i x_i = 0 \rightarrow \text{RESIDUAL AVERAGE ONTHOGOML. TO X IN THE SAMPLE.}$$

OLS DOES NOT SYSTEMATICALLY OVER/UNDER PREDICT Y IN THE SAMPLE

THERE IS NO REMAINING LINEAR ASSOCIATION BETWEEN X AND THE RESIDUALS.

$$\textcircled{4} \sum_{i=1}^n \hat{y}_i u_i = \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i) u_i$$

$$= \hat{\beta}_0 \sum_{i=1}^n u_i + \hat{\beta}_1 \sum_{i=1}^n x_i u_i$$

= 0. $\therefore$ RESIDUALS ARE ALSO ORTHOGONAL TO FITTED VALUES.

③ REGRESSION LINE PASSES THROUGH $(\bar{x}, \bar{y})$

$$\begin{aligned}\bar{\hat{y}} &= \frac{1}{n} \sum_{i=1}^n \hat{y}_i \\ &= \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 \cdot x_i) \\ &= \hat{\beta}_0 + \hat{\beta}_1 \cdot \bar{x}\end{aligned}$$

BUT NOTE THAT:

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \cdot \bar{x} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \cdot \bar{x} \\ \frac{1}{n} \cdot \sum_{i=1}^n \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \cdot \bar{x} \\ \frac{1}{n} \cdot n \cdot \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \cdot \bar{x} \\ \boxed{\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \cdot \bar{x}}\end{aligned}$$

$\therefore \bar{\hat{y}} = \bar{y}$ $\therefore$ WHEN X IS EQUAL ITS MEAN, THE LINE PREDICTS EXACTLY $\bar{y}$. $\therefore$ OLS REGRESSION LINE ALWAYS PASSES THROUGH THE POINT $(\bar{x}, \bar{y})$.

④ IN REGRESSION ANALYSIS WE WANT TO KNOW HOW MUCH OF THE TOTAL VARIATION IN y IS EXPLAINED BY THE MODEL $\Rightarrow$ WE WILL DECOMPOSE THE TOTAL VARIATION INTO TWO PARTS: ONE ATTRIBUTED TO THE FITTED LINE AND ONE TO THE RESIDUALS.

$$\begin{aligned}\text{TSS} &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ \text{ESS} &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \\ \text{RSS} &= \sum_{i=1}^n \hat{u}_i^2\end{aligned}$$

$$\begin{aligned}y_i &= \hat{y}_i \\ &= \hat{\beta}_0 + \hat{\beta}_1 \cdot x_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i + y_i \\ &= \hat{\beta}_0 + \hat{\beta}_1 \cdot x_i + (y_i - \hat{\beta}_0 - \hat{\beta}_1 \cdot x_i) \\ \boxed{y_i &= \hat{y}_i + \hat{u}_i}\end{aligned}$$

NOW

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{u}_i$$

SUM AND SQUARED

$$\begin{aligned}\sum_{i=1}^n (y_i - \bar{y})^2 &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n \hat{u}_i^2 + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y}) \hat{u}_i \\ \text{TSS} &= \text{ESS} + \text{RSS} + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y}) \hat{u}_i \quad \textcircled{1}\end{aligned}$$

BUT WE PROOF BEFORE THAT $\hat{u}_i$ IS ORTHOGONAL TO $\hat{y}_i$, THEN, $\textcircled{1}$ IS ZERO

$$\therefore \text{TSS} = \text{ESS} + \text{RSS}$$

$$\begin{aligned}\frac{\text{TSS}}{\text{TSS}} &= \frac{\text{ESS}}{\text{TSS}} + \frac{\text{RSS}}{\text{TSS}} \\ 1 &= R^2 + \frac{\text{RSS}}{\text{TSS}}\end{aligned}$$

$\therefore R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$ $\left. \begin{array}{l} \text{HOW MUCH OF} \\ \text{THE TOTAL} \\ \text{VARIATION IN DEPENDANT} \\ \text{VARIABLE IS EXPLAINED} \\ \text{BY THE MODEL.} \end{array} \right\}$

④ POPULATION REGRESSION FUNCTION

$$m(x) = E[Y | X=x]$$

① ASSUME THAT IS LINEAR, SHOWS THAT

$$E[u|x] = 0$$

$$E[Y | X=x] = \beta_0 + \beta_1 x$$

$$\text{SINCE } u = Y - \beta_0 - \beta_1 x$$

$$\begin{aligned} E[u|x] &= E[Y - \beta_0 - \beta_1 x | x] \\ &= E[\beta_0 + \beta_1 x - \beta_0 - \beta_1 x | x] \\ &= \beta_0 + \beta_1 x - \beta_0 - \beta_1 x \\ &= 0 \end{aligned}$$

→ AFTER CONTROLLING FOR X , THE REMAINING COMPONENT u HAS NO SYSTEMATIC IMPACT.

⑤ WITH IID ASSUMPTION AND ZERO CONDITIONAL MEAN, OLS ESTIMATOR IS UNBIASED.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$y_i - \bar{y} = \beta_1 (x_i - \bar{x}) + (u_i - \bar{u})$$

$$\begin{aligned} \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) &= \beta_1 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &\quad + \underbrace{\sum_{i=1}^n (x_i - \bar{x})(u_i - \bar{u})}_{(1)} \end{aligned}$$

$$\sum_{i=1}^n (x_i - \bar{x}) \bar{u} = \bar{u} \sum_{i=1}^n (x_i - \bar{x}) = 0$$

$$\therefore (1) = \sum_{i=1}^n (x_i - \bar{x}) \cdot u_i$$

$$\text{DEFINING } S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \text{ THEN}$$

$$\frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x}) u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_1 = \beta_1 + [2]$$

$$\hat{\beta}_1 - \beta_1 = [2]$$

$$\text{UNBIASED} \Rightarrow E[\hat{\beta}_1] = \beta_1$$

THEN

$$\begin{aligned} E[\hat{\beta}_1 - \beta_1 | x] &= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) \cdot E[u_i | x] \\ &= 0 \end{aligned}$$

$$\therefore E[\hat{\beta}_1 | x] = \beta_1$$

$$\therefore E[E[\hat{\beta}_1 | x]] = E[\beta_1] = \beta_1$$